

Eco-Intelligent Traffic Signal Control via Deep Reinforcement Learning

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Abstract—Traffic congestion remains a major challenge in modern urban areas, leading to delays, increased fuel consumption, and environmental pollution. Traditional traffic signal systems rely on fixed schedules and often fail to adapt to real-time traffic dynamics. This paper proposes an adaptive traffic signal control framework using Deep Reinforcement Learning (DRL), specifically the Proximal Policy Optimization (PPO) algorithm, to optimize signal phases based on real-time traffic conditions. The control problem is formulated as a Markov Decision Process, where the agent learns optimal actions from vehicle-level metrics such as waiting time, queue length, speed, and distance to the intersection. The system is trained and evaluated in a SUMO-based simulation environment and benchmarked against a conventional fixed-time controller. Experimental results show that the RL-based controller reduces average waiting time and queue lengths by up to 37.5% and 33.3%, respectively. It also achieves moderate improvements in speed, fuel consumption, and CO₂ emissions. These results highlight the potential of DRL for intelligent traffic management in smart city infrastructures.

Index Terms—Adaptive Systems, Proximal Policy Optimization, Reinforcement Learning, Smart Transportation, SUMO(Simulation urbaine de mobilité), Traffic Signal Control, Urban Mobility

I. INTRODUCTION

Traffic congestion is a persistent and growing problem in urban areas worldwide [1], leading to increased travel times, fuel consumption, and air pollution [2], [3]. Traditional traffic signal control systems typically operate on fixed-time schedules or simple reactive logic, which often fail to adapt effectively to dynamic and unpredictable traffic patterns [4], [5]. As urban populations grow and transportation networks become more complex, there is a critical need for intelligent traffic management solutions that can respond in real-time to changing road conditions [6], [7].

Recent advancements in artificial intelligence, particularly in Reinforcement Learning (RL), offer promising opportunities to improve traffic control through adaptive, data-driven

decision-making [8], [9]. Reinforcement learning enables an agent to learn optimal control strategies by interacting with the environment and receiving feedback in the form of rewards [10]. Unlike rule-based or pre-timed systems, RL-based controllers can evolve their behavior over time and adapt to diverse traffic scenarios without explicit programming [11].

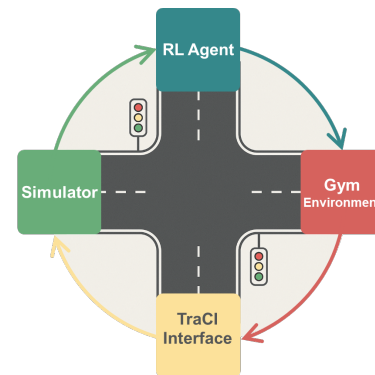


Fig. 1. Illustration of the traffic control framework and RL agent interaction.

This paper presents an RL-based approach for optimizing traffic signal control at road junctions using a simulation environment built on SUMO (Simulation of Urban Mobility) [12]. We model the traffic light control task as a Markov Decision Process and train a deep reinforcement learning agent using the Proximal Policy Optimization (PPO) algorithm [13]. The agent observes real-time traffic metrics such as vehicle waiting time, queue length, speed, and distance to the intersection and selects appropriate signal phases to minimize overall congestion.

The proposed system is structured around four main components that form a closed learning loop: the RL agent, the TraCI interface, and the SUMO traffic simulator. These components work in coordination to allow real-time interaction between traffic conditions and signal control decisions, enabling the

agent to observe, learn, and act continuously. This architectural layout is shown in Figure 1, providing a high-level overview of how each element contributes to the learning and control process.

To evaluate the performance of the proposed method, we compare it against a traditional fixed-time signal control system across several key metrics: average waiting vehicles, vehicle speed, queue length, CO₂ emissions, and fuel consumption. Simulation results demonstrate that the RL-based controller significantly improves traffic efficiency while showing potential environmental benefits.

This study addresses adaptive traffic signal control through the following key components:

- Designing a realistic reinforcement learning environment for traffic light control by integrating SUMO with a customized simulation interface.
- Proposing a compact yet informative state representation based on vehicle-level metrics from each incoming road.
- Applying the Proximal Policy Optimization (PPO) algorithm to train an adaptive traffic signal agent capable of real-time decision-making.
- Providing comprehensive evaluation results that demonstrate improvements over fixed-time control, including reductions in delay, emissions, and fuel consumption.

The remainder of this paper is structured as follows. Section II provides a review of related work in adaptive traffic signal control using machine learning techniques. Section III details the proposed system architecture, reinforcement learning formulation, and simulation environment. Section IV presents the experimental results and performance comparisons between the RL-based controller and the traditional fixed-time approach. Section V discusses the limitations of the current study and potential avenues for future research. Finally, Section VI concludes the paper by summarizing the key findings and contributions.

II. RELATED WORK

Recent advances in artificial intelligence (AI) have spurred significant research into adaptive traffic signal control systems. Approaches based on reinforcement learning, supervised learning, heuristic optimization, and fuzzy logic have all been applied to both single-junction and network-level traffic management systems. This section summarizes key contributions from 2020 onward.

A. Reinforcement Learning

RL-based traffic signal control has shown promising results for both isolated intersections and networked environments. Single-agent deep RL models such as Deep Q-Networks (DQN) have demonstrated substantial improvements over fixed-time control by learning adaptive phase timings. Multi-agent RL methods extend this by enabling coordination across intersections. For example, a cooperative multi-intersection Q-learning method achieved a 13% reduction in travel time and an 11% decrease in fuel usage compared to static policies [14]. Similarly, federated multi-agent approaches using PPO [15]

improved convergence speed and reduced vehicle delay by up to 27%. However, RL can be sensitive to extreme traffic congestion, and training under highly saturated scenarios may lead to suboptimal generalization [16]. Despite these challenges, RL remains one of the most adaptive and scalable methods for dynamic traffic environments.

B. Supervised Learning

Supervised learning has also been used to develop traffic signal controllers, often leveraging existing datasets or optimized traffic patterns. A neural network trained with just a few expert-optimized examples outperformed conventional greedy heuristics in managing complex traffic networks [17]. Hybrid methods that combine supervised learning with reinforcement learning have been proposed to improve sample efficiency and training stability. For instance, initializing an RL agent with supervised demonstrations before fine-tuning with RL has shown to reduce training time and improve early performance. While pure supervised learning lacks adaptability to unforeseen conditions, it remains a valuable strategy for bootstrapping intelligent traffic controllers.

C. Heuristic Optimization

Heuristic and metaheuristic algorithms, such as Genetic Algorithms (GA) [18], Particle Swarm Optimization (PSO), and Ant Colony Optimization (ACO), have long been applied to optimize traffic signal timing. A recent PSO-based controller reduced queue lengths by 20.4% and waiting times by 17.9% at a busy junction compared to fixed-timing plans [19]. Multi-objective evolutionary algorithms like NSGA-II have also been used to generate Pareto-optimal solutions that balance delay, emissions, and pedestrian wait times. These methods excel in offline optimization but may struggle with real-time adaptability, making them suitable for planning tasks or as components of hybrid systems.

D. Fuzzy Logic

Fuzzy logic controllers remain relevant for traffic control due to their interpretability and robustness to uncertain inputs. A fuzzy controller using vehicle-to-infrastructure (V2I) communication reduced average vehicle waiting time from 25 seconds to around 21–22 seconds in a recent 2023 study [20]. Hybrid models that combine fuzzy inference with RL—such as fuzzy-state deep Q-networks—have also been introduced, improving waiting times by up to 18% over traditional DQN approaches [21]. While fuzzy logic excels in rule-based reasoning, its scalability to large networks and adaptability to unforeseen patterns remain limited unless enhanced by learning algorithms.

III. METHODOLOGY

This section describes the design and implementation of our RL-based traffic light control system. We begin with an overview of the environment and system integration, followed by a detailed explanation of the reinforcement learning setup.

A. System Architecture Overview

Our approach integrates the Simulation of Urban MObility (SUMO) traffic simulator with a custom environment to train a reinforcement learning agent capable of adaptive traffic light control. The agent interacts with the simulation in a closed-loop manner, observing traffic conditions and selecting the appropriate signal phase at each timestep. This setup involves several interconnected modules—such as traffic data collection, action translation, and feedback processing—that enable seamless coordination between the agent and the simulation. The overall structure and data flow within the system are illustrated conceptually in Figure 2, highlighting the key components supporting real-time learning and control.

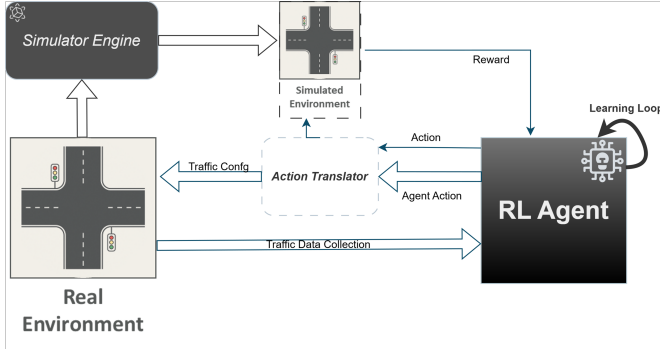


Fig. 2. Architecture of the PPO-Based Adaptive Traffic Signal Control Framework

As shown in the system architecture diagram, the proposed framework establishes a continuous interaction loop between the RL agent and the traffic environment. At the core of this interaction is the Action Translator, which bridges the agent's discrete action outputs with the configuration commands required by the SUMO simulator. The simulator engine generates traffic scenarios and updates the environment state, which is monitored through virtual sensors to collect real-time traffic data such as queue lengths and waiting times. This information is processed and fed back into the RL agent as part of the observation space, allowing the agent to adapt its decisions over time based on feedback in the form of rewards. Additionally, the architecture is designed to support integration with real-world traffic data sources, enabling smooth transitions between simulated training and practical deployment.

B. Reinforcement Learning for Traffic Light Control

We formulate the traffic light control problem as a Markov Decision Process (MDP), which is solved using model-free reinforcement learning. The goal of the agent is to learn an optimal policy that minimizes overall vehicle waiting time and improves traffic flow.

Figure 3 outlines the general decision-making loop in a reinforcement learning-based traffic signal control system. It highlights how the RL agent observes various traffic metrics from the environment—such as vehicle count, queue length,

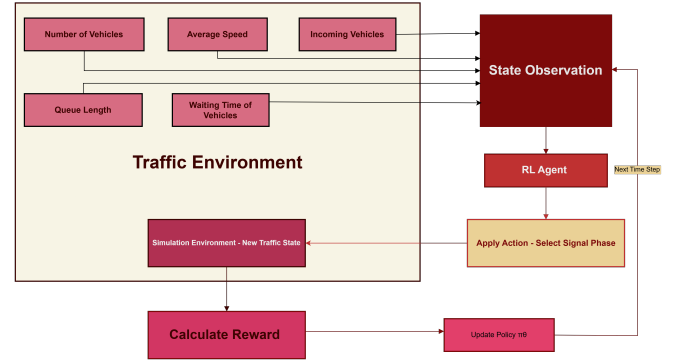


Fig. 3. RL-based decision loop for traffic signal optimization.

and speed—then selects a traffic signal phase based on this state, receives feedback in the form of a reward, and updates its policy accordingly. This closed-loop interaction allows the agent to learn and refine its decisions over time. Building on this foundation, we now describe how the traffic control task is formally framed as a Markov Decision Process (MDP) and detail the components involved in the learning cycle.

1) *State Representation*: At each decision point, the agent receives a state vector representing real-time traffic data from the four incoming edges at the intersection. As illustrated in the traffic environment component of Figure 3, this state is composed of key metrics that describe the current conditions on the road. For each edge, the following four metrics are collected:

- W_i : Total waiting time of vehicles.
- N_i : Number of vehicles.
- v_i : Average speed of vehicles.
- d_i : Average distance of vehicles from the traffic light.

The resulting **16-dimensional state vector** is:

$$s_t = [W_1, N_1, v_1, d_1, \dots, W_4, N_4, v_4, d_4]$$

This encoding provides the agent with spatial and temporal awareness of traffic conditions.

2) *Action Space*: The action space is discrete with four possible actions, each corresponding to granting a green light to one of the four incoming roads while keeping others red. Formally:

$$a_t \in \{0, 1, 2, 3\}$$

where each action maps to a phase configuration of the traffic signal (e.g., $a = 0$ gives green to East, red to others).

3) *Reward Function*: The reward function is defined to minimize total waiting time:

$$r_t = - \sum_{i=1}^4 W_i$$

Alternative formulations including emissions, queue length, or fuel consumption were considered and can be incorporated for multi-objective optimization.

4) *Learning Algorithm:* We use PPO from `stable-baselines3`, a reliable on-policy algorithm that balances exploration and policy stability. The PPO agent is trained over 100,000 timesteps using a multilayer perceptron (MLP) policy network.

The agent updates its policy $\pi_\theta(a|s)$ to maximize expected cumulative discounted rewards:

$$\mathbb{E}_{\pi_\theta} \left[\sum_{t=0}^T \gamma^t r_t \right]$$

Where:

- γ : discount factor (typically 0.99),
- T : episode horizon (1000 steps),
- θ : network parameters.

C. Simulation Environment

The simulation environment consists of a small synthetic traffic network built with the SUMO simulator. It contains **three intersections**, each regulated by traffic lights and identified as nodes L, R, and U. These intersections are connected by **12 major roads**, with additional links to surrounding regions (e.g., East, West, SouthEast, SouthWest, NorthEast, and NorthWest). Each road typically includes **three lanes per direction**, supporting straight, left-turn, and right-turn movements. The network is composed mainly of multi-lane straight segments and connector roads for intersection turns.

Traffic flows dynamically across the intersections, and the environment captures realistic traffic behaviors such as vehicle queuing, acceleration, and deceleration. Static signal control logic is initially defined, but it is overridden during RL-based training to allow adaptive, real-time signal phase selection. Vehicles entering the network are generated from multiple boundary points, ensuring diverse and variable traffic patterns for the learning agent.

D. Algorithm Implementation

The training process of the PPO-based traffic light control agent is outlined in the algorithm. The agent interacts with the simulation environment and collects experiences and updates its policy using the PPO algorithm.

[H] PPO Training for Traffic Light Control

Environment env , total timesteps N , batch size B , learning rate α , discount factor γ , policy π_θ Optimized policy π_θ

Initialize policy network π_θ and value network V_ϕ Initialize empty rollout buffer $\mathcal{D}$

$t = 1$ N

Observe state s_t Sample action $a_t \sim \pi_\theta(a_t|s_t)$ Execute action a_t in env Receive reward r_t , next state s_{t+1} , and done flag Store transition $(s_t, a_t, r_t, s_{t+1}, done)$ in $\mathcal{D}$

size of $\mathcal{D} \geq B$

transition in $\mathcal{D}$ Estimate value $V_\phi(s_t)$ Compute advantage $\hat{A}_t$ and target return R_t

$k = 1$ K epochs Sample minibatches from $\mathcal{D}$ Update π_θ using PPO loss:

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)\hat{A}_t) \right]$$

Update V_ϕ by minimizing value loss $\|V_\phi(s_t) - R_t\|^2$
Clear buffer $\mathcal{D}$

IV. RESULTS AND DISCUSSION

In this section, we evaluate the performance of our RL-based traffic light control system against a baseline Fixed Time control strategy. The comparison is based on key traffic efficiency and environmental metrics recorded over 1000 simulation steps. These include average waiting vehicles, vehicle speed, queue length, CO₂ emissions, and fuel consumption.

A. Average Waiting Vehicles

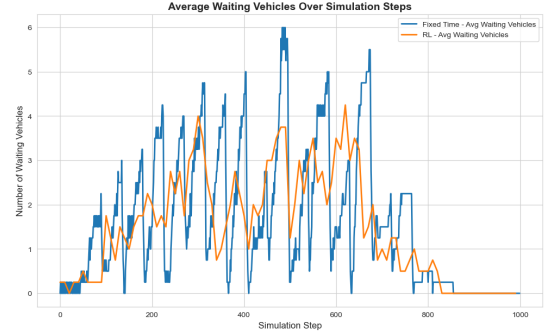


Fig. 4. Average Waiting Vehicles Over Simulation Steps

Figure 4 shows that the RL agent consistently achieves a lower number of average waiting vehicles compared to the fixed-time controller, especially during peak traffic periods (steps 200–700). The RL policy dynamically adapts to traffic conditions, reducing unnecessary red light durations and preventing queue buildup. In contrast, the fixed-time controller often leads to longer queues due to its static nature.

RL reduces average waiting vehicles, particularly under dynamic or non-uniform traffic loads, validating its adaptability.

B. Average Speed

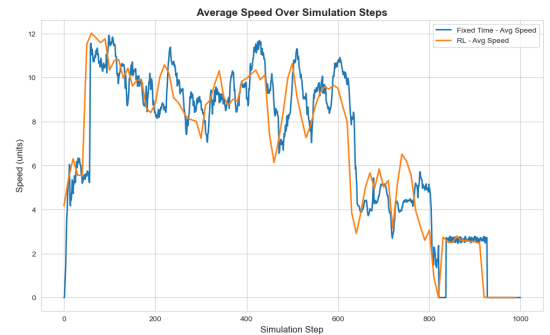


Fig. 5. Average Speed Over Simulation Steps

In Figure 5, we observe that the RL-controlled system generally maintains higher average vehicle speeds compared to the fixed-time setup. This is a direct consequence of reduced queuing and smoother flow, allowing vehicles to move more freely through the intersection.

RL enables smoother flow and maintains higher speeds, especially during congestion peaks, thereby enhancing overall traffic throughput.

C. Queue Length

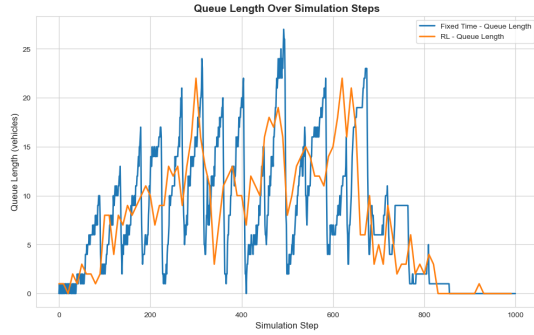


Fig. 6. Queue Length Over Simulation Steps

As shown in Figure 6, the queue length under the fixed-time policy frequently spikes and remains volatile throughout the simulation. On the other hand, the RL policy stabilizes queue length and reduces it more efficiently as the episode progresses. Notably, after the peak congestion period, the RL policy facilitates quicker dissipation of traffic queues.

Reinforcement learning enables better queue management and faster recovery after congestion peaks.

D. CO₂ Emissions

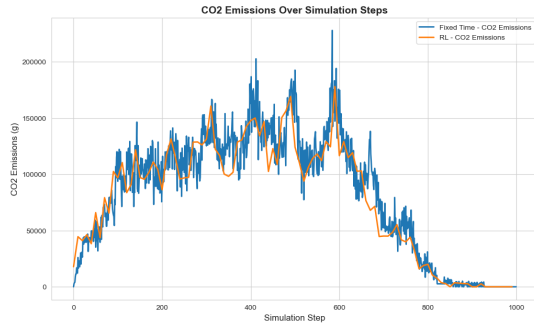


Fig. 7. CO₂ Emissions Over Simulation Steps

In Figure 7, we compare CO₂ emissions generated under both control policies. The RL-based traffic signal controller produces smoother and generally lower emission curves compared to the fixed-time strategy. This improvement results from the model's effectiveness in reducing queue length and vehicle waiting time, which leads to a clearer intersection and allows vehicles to move through more consistently. As shown in Figure 5, this contributes to a more stable average speed profile, reducing the need for frequent stopping and acceleration—key contributors to excess fuel consumption and emissions.

However, the observed reduction in CO₂ emissions remains modest. This is because CO₂ output is influenced not only

by queueing and delays but also by other factors such as engine type, acceleration patterns, and traffic heterogeneity, which are not directly controlled or modeled by the RL agent. Therefore, while the agent clearly improves traffic flow, its impact on emissions is an indirect benefit rather than a primary optimization target.

E. Fuel Consumption

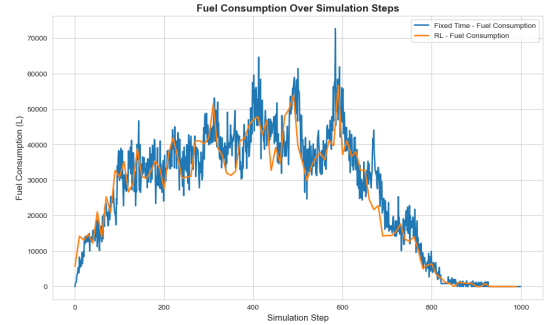


Fig. 8. Fuel Consumption Over Simulation Steps

Fuel consumption patterns in Figure 8 follow similar trends to CO₂ emissions. The RL policy shows a moderate reduction in fuel usage, especially in the second half of the simulation, which aligns with the improved queue management and smoother vehicle flow observed earlier.

RL contributes to fuel efficiency, supporting its utility not only for traffic performance but also for environmental impact.

F. Summary of Improvements

TABLE I
PERCENTAGE CHANGE IN PERFORMANCE INDICATORS AND OPTIMIZATION EVALUATION

Indicator	Mean Change (%)	Max Change (%)
Average Queue Length	11.41%	33.33%
Average Waiting Time (W_i)	15.92%	37.50%
CO ₂ Emissions (g)	6.93%	24.04%
Fuel Consumption (mL)	6.93%	24.04%

The results demonstrate that reinforcement learning can substantially enhance traffic signal efficiency compared to traditional fixed-time control. The most prominent improvements are seen in waiting time and queue length—two of the most critical indicators of congestion. While environmental benefits (emissions and fuel) are more subtle, they still show positive trends attributable to smoother traffic flow.

These outcomes validate the use of real-time vehicle-level state information and show that a policy trained in a simplified intersection can still produce generalizable behaviors under varying traffic patterns. Future work can extend this to larger, multi-intersection networks and explore multi-objective reward functions that explicitly weigh environmental metrics alongside traffic efficiency.

G. Work Limitations

Despite its effectiveness, the proposed approach has several limitations. First, reinforcement learning typically requires significant computational resources and training time compared to traditional optimization algorithms such as rule-based systems or heuristic scheduling. This is due to the need for high-volume simulations and iterative policy updates, especially when using deep neural networks as function approximators. Second, although simulation-based training enables safe and efficient development, the learned policy may struggle to generalize to real-world traffic scenarios that involve rare or unpredictable events—such as accidents, emergency vehicles, or infrastructure failures—which are not easily modeled or represented in synthetic environments. Lastly, the current model assumes perfect, noise-free traffic data and does not account for latency or sensor uncertainty, which are common in real-world deployments. Addressing these limitations will be crucial for transitioning from simulation to deployment in practical, large-scale urban settings.

V. CONCLUSION AND FUTURE WORK

This work proposed a RL-based approach for adaptive traffic light control, utilizing real-time vehicle-level state information within a SUMO simulation environment. By modeling the intersection as a Markov Decision Process and training a PPO agent, the system learned to dynamically select traffic signal phases in response to observed traffic conditions.

Experimental results demonstrated that the RL-controlled system significantly outperformed traditional fixed-time control across multiple performance metrics. Notably, the RL agent reduced average vehicle waiting time and queue length, maintained higher average speeds during congestion, and achieved moderate reductions in fuel consumption and CO₂ emissions. These outcomes highlight the adaptability and efficiency of learning-based methods for traffic control in dynamic environments.

While promising, this research serves as a foundation for several future extensions. One important direction is the co-ordination of multiple intersections using multi-agent reinforcement learning to enable city-scale traffic optimization. Additionally, incorporating multi-objective reward functions can help balance trade-offs between traffic efficiency, environmental sustainability, and fairness. Further work will also explore the generalization of trained policies to new intersection layouts and traffic patterns, potentially using transfer learning. Finally, adapting this framework for deployment in real-world smart city infrastructure, leveraging real-time sensor data, will be a critical step toward practical application.

In summary, this study demonstrates the potential of reinforcement learning as a flexible and effective strategy for intelligent traffic signal control, while also identifying key areas for future research and improvement.

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